

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 items, 1 point each — 8 points)

Write the letter of the best definition on the line beside each term. **Two definitions will not be used.**

- | | |
|---------------------------------------|--------------------------------|
| _____ 1. Greatest common factor (GCF) | _____ 5. Zero product property |
| _____ 2. Prime polynomial | _____ 6. Double root |
| _____ 3. Perfect square trinomial | _____ 7. Excluded value |
| _____ 4. Difference of squares | _____ 8. Rational expression |

- (A) A quotient of two polynomials.
(B) A binomial of the form $a^2 - b^2$; it factors as $(a + b)(a - b)$.
(C) The largest factor shared by every term of a polynomial.
(D) A number that makes a denominator zero, so the expression is undefined there.
(E) An expression written as a product of two or more expressions.
(F) A polynomial with integer coefficients that cannot be factored any further over the integers.
(G) If a product equals zero, then at least one of its factors equals zero.
(H) A solution that appears twice because the factor producing it is squared.
(I) The exponent that tells how many times a base is used as a factor.
(J) A trinomial of the form $a^2 \pm 2ab + b^2$; it factors as $(a \pm b)^2$.

Part B — Multiple Choice (6 items, 2 points each — 12 points)

Circle the best answer.

- Which expression is equal to $(2m + 3)^2$?
(A) $4m^2 + 9$
(B) $4m^2 + 6m + 9$
(C) $4m^2 + 12m + 9$
(D) $2m^2 + 12m + 9$
- What is the value of $\sqrt{9 + 16}$?
(A) 7
(B) 5
(C) 25
(D) 12
- Which is the **complete** factorization of $6x^2 + 9x$?
(A) $x(6x + 9)$
(B) $3(2x^2 + 3x)$
(C) $3x(2x + 3)$
(D) $3x(2x^2 + 3)$

4. Which polynomial is **prime** over the integers?
- (A) $x^2 - 49$
 - (B) $x^2 + 49$
 - (C) $x^2 - 14x + 49$
 - (D) $x^2 + 14x + 49$
5. How many solutions does $x^3 + 9x = 0$ have, and of what type?
- (A) 3 solutions: 3 real
 - (B) 3 solutions: 1 real and 2 imaginary
 - (C) 1 solution: 1 real
 - (D) 2 solutions: 2 real
6. Which simplification is **correct**?
- (A) $\frac{x+5}{x+7} = \frac{5}{7}$
 - (B) $\frac{x+5}{5} = x$
 - (C) $\frac{5(x+5)}{5(x+7)} = \frac{x+5}{x+7}$
 - (D) $\frac{x^2+5}{x^2} = 5$

Part C — Short Answer & Computation (10 items, 5 points each — 50 points)

Show all work. An answer with no supporting work earns no credit.

1. Simplify completely. Write your answer with positive exponents only. $\frac{(3x^2y^3)^2 \cdot 2xy}{6x^3y^4}$
2. (a) Simplify $\sqrt{75x^3}$, where $x > 0$. (b) Write $\sqrt[3]{x^5}$ using a rational exponent. (c) Evaluate $16^{3/4}$.
3. Factor completely by grouping: $3x^3 + 12x^2 + 2x + 8$
4. Factor completely: $4x^2 + 11x + 6$

5. Factor completely. (a) $49x^2 - 64$ (b) $x^3 - 27$
6. Solve by factoring: $2x^2 = 5x + 12$
7. Simplify, and state **every** excluded value. $\frac{x^2 - 16}{x^2 - x - 12}$
8. Divide and simplify, and state **every** excluded value. $\frac{x^2 - 4}{x^2 + 6x + 9} \div \frac{x + 2}{x + 3}$
9. A rectangular banner has area $A(x) = 2x^2 + 11x + 12$ square feet and width $x + 4$ feet. (a) Find an expression for its length. (b) Find the length, width, and area when $x = 5$.
10. Factor completely. (a) $5x^3 - 45x$ (b) $2x^4 - 32$

Part D — Extended Response (2 items, 15 points each — 30 points)

1. **Verify, then judge a claim.**

(a) Verify the identity $(x - 4)(x^2 + 4x + 16) = x^3 - 64$ by multiplying.

- (b) A student writes: “ $\frac{x^3 - 64}{x - 4}$ simplifies to $x^2 + 4x + 16$, *for every value of x .*” Test both expressions at $x = 5$ and at $x = 4$.

- (c) Is the student’s algebra correct? Is the student’s *claim* correct? Explain in one or two sentences, using the word **excluded**.

2. **A rocket is launched from a platform.** Its height in feet after t seconds is $h(t) = -16t^2 + 64t + 80$.

- (a) Factor $h(t)$ completely and find both values of t for which $h(t) = 0$.

- (b) Which of those two values describes when the rocket lands? What does the other value mean here, and why is it rejected?

- (c) Solve $h(t) = 80$. Interpret **both** solutions — this time neither one is rejected.

- (d) How many solutions does the equation $h(t) = 0$ have, and of what type? How does the degree of $h(t)$ tell you this before you factor anything?
